

5a. Pros and cons of each allocation method

Array of arrays (separately malloc'd rows):

Pros:

Easy to understand.

Each row can be allocated, resized.

Each row can have different lengths.

Cons:

Requires many small allocations.

Rows are not contiguous in memory.

Need an extra pointer to the element.

One big contiguous block (row pointers into a single block):

Pros:

All data in one contiguous memory block.

Fewer allocations.

Cons:

Can't resize or free individual rows separately.

Need more setup required to align pointers with row starts.

Less flexible if row lengths vary.

5b. Situations where one is better than the other

Array of arrays is better when you need flexibility.

One big contiguous block is more suitable for users who prioritize cache efficiency and quick browsing.